



REPRODUCTION & EMBRYO DEVELOPMENT — FULL MIND MAP

Biology Ch. 7.ix · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this sub-chapter:

1. Fertilization — Site in Human Female Reproductive Tract
2. Sperm Fertilizing Capacity
3. Human Embryo — Beginning of Heartbeat
4. Amnion — Protection of Developing Embryo
5. Sertoli Cells — Testis & Spermatogenesis

1. Fertilization — Site in Human Female Reproductive Tract

- Human fertilization source ke according fallopian tube/oviduct mein hota hai—uterus se door, ovary ke close end par. Oviduct ovary ko uterus se connect karta hai aur ovulated oocyte isi route se travel karta hai. [Q1]
- Normally sperm ovum ko Fallopian tube mein fertilize karta hai. Source Fallopian tube ko oviduct/uterine tube bhi kehta hai aur ise sperm ko egg tak le jaane, fertilization ke liye suitable environment dene aur egg transport karne wali paired narrow ducts ke roop mein explain karta hai. [Q2]

2. Sperm Fertilizing Capacity

- Female genital tract/oviduct mein pahunchne ke baad sperm source ke mutabik lagbhag 72 hours tak alive reh sakta hai, lekin ovum ko fertilize karne ki capacity lagbhag 48 hours yani 1–2 days tak rehti hai. [Q3]



EXAM TRAP — Sperm Life vs Fertilizing Capacity

- ▶ Source wording yaad rakho: sperm survival ≈ 72 hours, but fertilizing capacity ≈ 48 hours (2 days). [Q3]

3. Human Embryo — Beginning of Heartbeat

- Human embryo ka heart source ke according fertilization ke about 4–5 weeks baad pehli baar beat karna start karta hai; given options mein Fourth week correct marked hai. Is stage par heartbeat roughly 105–121 beats per minute batayi gayi hai. [Q4]

4. Amnion — Protection of Developing Embryo

- Amnion embryo ko cover karne wali membrane hai. Amniotic fluid se filled hokar amniotic sac banata hai aur developing embryo ko desiccation (sukhne), outer pressure etc. se protective environment deta hai. [Q5]

5. Sertoli Cells — Testis & Spermatogenesis

- Sertoli cells human testes mein milne wali somatic/supporting cells hain. Ye seminiferous tubules ka structural component hote hue spermatogenesis mein contribute karti hain aur source inhe sustentacular “nurse” cells bhi kehta hai. [Q6]
- Source ke mutabik Sertoli cells adenohypophysis se secreted FSH se activated hoti hain aur apni membrane par FSH receptors express karti hain. [Q6]